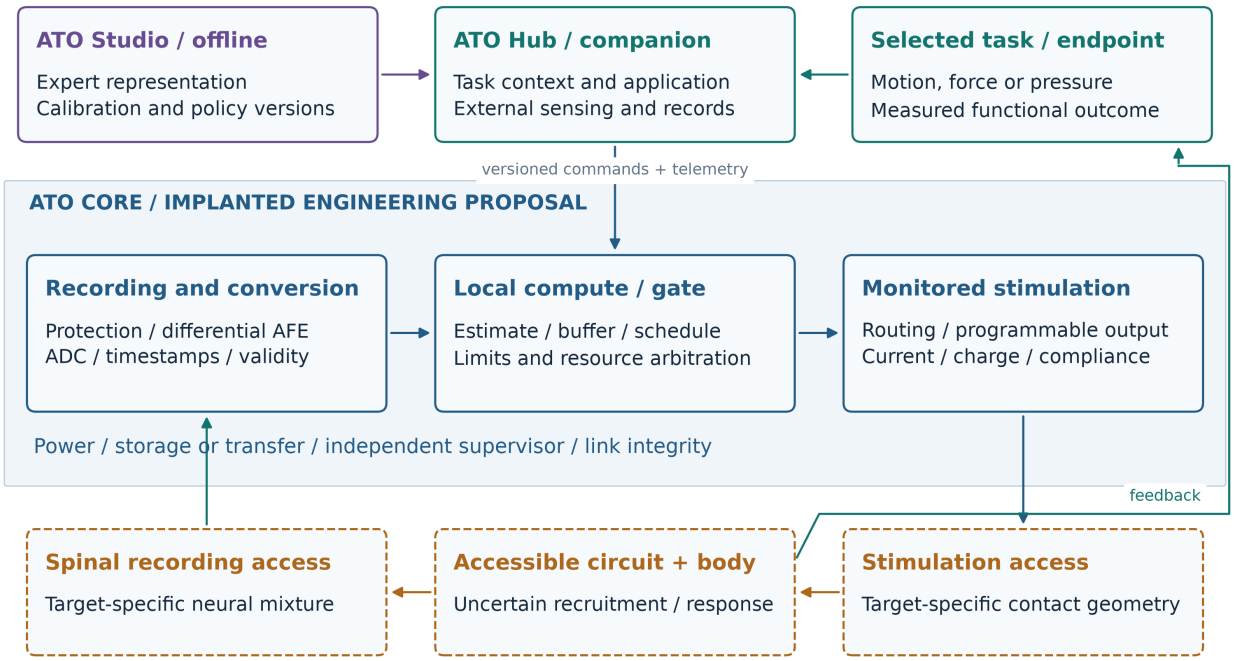
From spinal observations to an admissible intervention

A digital evidence report for expert-informed practice and selected physiological feedback

Pratham · Project ATO · 12 October 2026



Abstract. Project ATO proposes implanted spinal recording, computational interpretation, individualized stimulation and measured feedback. This revision tests intermediate requirements rather than presuming an ideal decoder or replaying expert voltages. New digital experiments separate teacher competence from reference fidelity, evaluate moving sources at contact apertures, validate causal forecasts, identify finite pulse responses, and derive prospective monitor precision. A connected reset-trial pilot withholds every intervention under the tested learner drift. Those failures define the next engineering choices.

Evidence level. All new measurements in this report are generated by declared artificial electrical/body/source models. No new human, animal or physical-device experiment is reported. Empirical allocations are finite model checks; they are not biological error guarantees.

AstroSam / Mind-Link inspires the progression from a future concept to inspectable engineering. The full native electronics, populated CAD and integrated final manuscript remain later work. This report covers selected Stage 3 revisions, not completion of the implant.

01 / Evidence becomes a design decision

The 28-page source paper is a research map. Its examples support individual mechanisms; the complete expert-spinal-recording → personalized intervention → retained learner benefit remains a conjunction to investigate. Primary reports [1–5] inform the controls below.

Experimental precedent	Design consequence	Limit retained
El Hadwe et al. [1]: short-term circumferential spinal recordings; movement decoding in five rats	Use a spatial contact model; compare neural features with measured movement; test sampling and ambiguity.	Movement correlation is not an expert command. Offline bidirectional processing is not causal online decoding.
McPherson et al. [2]: EMG-linked intraspinal stimulation during injured-rat retraining, with OFF follow-up	Match practice exposure; distinguish assistance from retained performance; retain contingent and timing controls.	Recovery with muscle triggering does not demonstrate expert neural transfer or doubled healthy learning speed.
Nishimura et al. [3]: timing-dependent cortical-spike-triggered spinal stimulation in two monkeys	Preserve processing, observation and onset delay; include nonbeneficial and harmful plasticity alternatives.	A hybrid brain–spine mechanism supplies no universal spinal learning rule.
Phillips et al. [4]: targeted lower-thoracic stimulation with measured pressure feedback after SCI	Keep target-specific pressor authority, external pressure sensing, disturbances, saturation and fallback tests.	No universal autonomic control, neural pressure decoder or healthy-user enhancement is established.
Formento et al. [5]: stimulation can interfere with proprioceptive signaling	Account for sensory cost and recording loss alongside immediate assistance.	A constant synthetic sensory penalty is a hypothesis, not a fitted human dose law.

The wider platform vision retains bladder, cough, sensory and pelvic functions in the capability map. Each needs its own accessible pathways, state measurement, input directions and endpoint. Triggering a response alone does not establish selective delay or sustained control.

02 / Qualify what the teacher supplies

The new teacher revision generates sixteen complete 12-second recordings: two disclosed artificial performers, two cadences and four observation-noise realizations. Each generator supplies 12 seconds for training and 84 seconds of withheld cadence/noise exposure. There is one artificial expert/contact geometry, not seven independent people.

Performer / cadence	Task competence	External reference	Neural reference
Original voluntary / 1 Hz	0 / 4	Unavailable	Unavailable
Original voluntary / 1.15 Hz	0 / 4	Unavailable	Unavailable
Separate inverse-model / 1 Hz	4 / 4	Finite-domain pass	Finite-domain pass
Separate inverse-model / 1.15 Hz	4 / 4	Unavailable	Unavailable

Competence does not guarantee representation fidelity. The faster inverse performer passes the contact-peak task criteria, while its frozen template fails whole-trajectory allocation. The inverse generator uses a private body time constant to produce competence; the fitted library receives only measured contact features and external positions.

Independent comparators. An unavailable neural decoder must not disqualify a competent external teacher reference. External and neural alternatives use equal capture exposure. Neural fitting uses measured external labels, so incremental neural value still needs a matched comparison. The result is a task representation for learner-specific control, not expert memory or voltage transfer.

Moving sources at the actual contact apertures

A separate 75-capture study evaluates the source at 4 kHz per contact with 3 microsecond contact skew. Six reference-grid comparisons retain paired noise and observation support. Holding moving source state for 5 ms fails 12 of 18 informative feature comparisons and six prediction comparisons under the fixed allocations. Dense-active neural packets remain unqualified. The fresh teacher captures therefore use actual apertures.

A deliberately ambiguous field produces identical contact features with conflicting measured positions. The observer must abstain in that declared construction. This is an observability witness, not proof that every biological spinal interface is ambiguous.

03 / A compact map of the revision

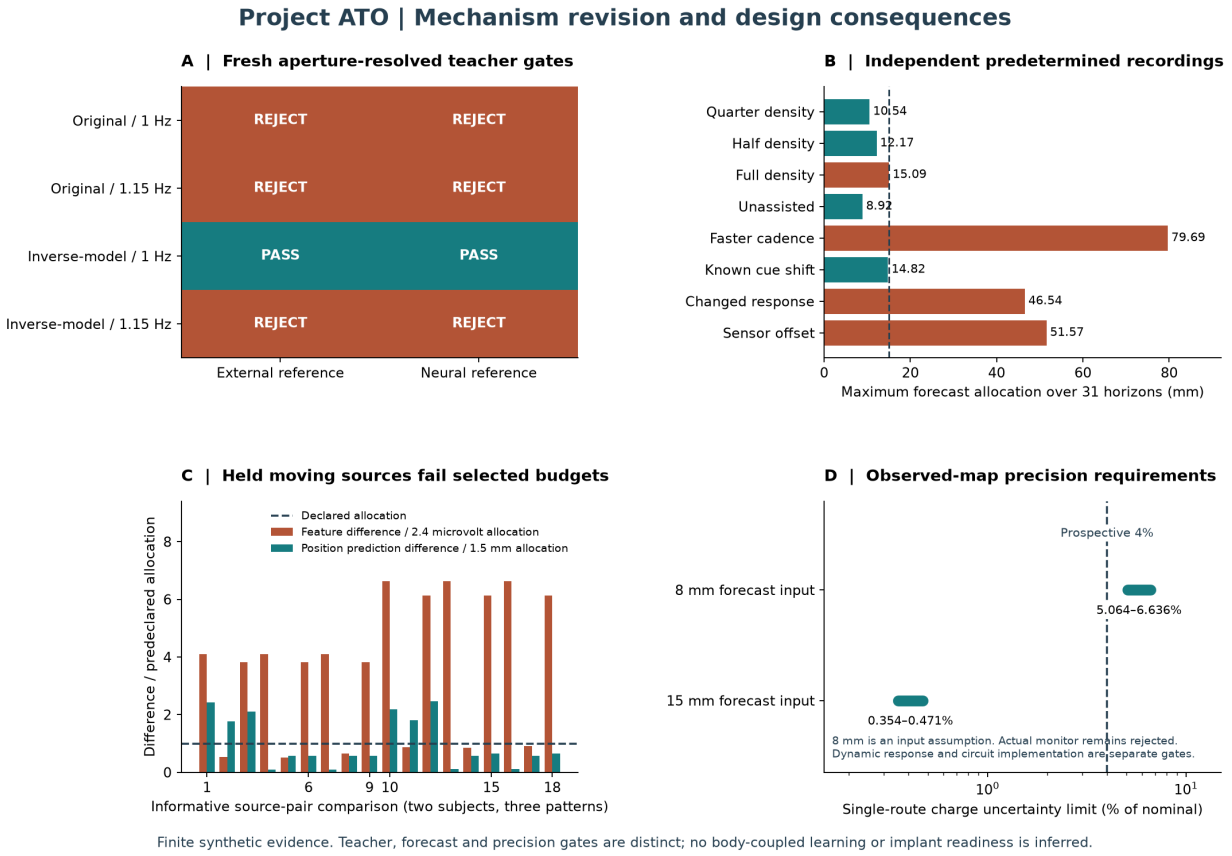


Figure 1. Four independent gates: teacher competence/reference availability, complete withheld forecast contexts, moving-source fidelity ratios and prospective delivery precision. A pass in one panel cannot replace a failed gate in another. The predictor keeps the 15 mm threshold, including the narrow full-density failure.

Revision	Preserved evidence
Teacher	Sixteen historical inputs remain unchanged; sixteen new aperture-resolved captures have separate identities.
Source	75 captures, 14,399,360 acquired contact samples and six grid pairs; absent/redundant controls remain distinct.
Forecast	31 fitted horizons and eight full withheld recordings; 248 recording/horizon checks, with four context passes and four rejects.
Precision	Four observed calibrations, 144 requirement rows and 24 thresholds; current-monitor rejection is retained.

04 / Predict without future measurements

The intervention-aware observer uses sixteen past external position samples, known task cues and a dose-history/declared-command grid. Complete monitor records become available after the 500 microsecond biphasic drive. Future measured charge, achieved movement, private activation, recruitment and skill coefficients are excluded from online inputs.

Four six-second predetermined command recordings train independent ridge models at every 5 ms horizon from 0 through 150 ms. Chronological fitting is purged. Eight complete new recordings test every horizon without refitting failed contexts.

Complete withheld recording	Largest allocation	15 mm gate
Quarter density	10.539 mm	Pass
Half density	12.175 mm	Pass
Full density	15.086 mm	Reject
Unassisted	8.925 mm	Pass
Known 15% faster cadence	79.689 mm	Reject
Known 125 ms cue shift	14.817 mm	Pass
Changed response	46.540 mm	Reject
Unannounced 40 mm sensor offset	51.571 mm	Reject

Runtime uncertainty remains live

Let R be the larger saved training/context residual plus the 3 mm label allocation, P the current dose-interval projection, and A the saved total allocation. The live allowance is $\max(A_{train}, A_{context}, R + P)$. Enlarging a monitor interval can reject an unchanged mean prediction. Saved interval maxima are not added twice.

Predetermined recordings do not qualify arbitrary feedback commands. For a proposed action, use a zero-future-command baseline that retains ongoing past response, then add the new candidate response once. For gain G and dose u, include $U_G|u| + |G|U_u + U_GU_u$, plus baseline and independent response residual allocations.

The current observer uses external learner measurements. Invalid spinal packets stay invalid; this result does not establish neural incremental value or retained learning.

05 / A pulse needs its own response model

The new dynamic experiment uses the unchanged electrical adapter for each nonzero pulse. The private body receives returned true charge; identification receives only completed independent monitor records and external measured positions. Twenty-four isolated signed calibration pulses are each paired with a separately measured resting baseline.

A 0–600 ms response table is fitted at 5 ms physical supports. Independent moving single, paired, repeated and changed-context trajectories are withheld. Integration at 2.5 ms and 0.625 ms is compared. Calibration gain intervals preserve the approximately 44.1% actual charge uncertainty; future 7.4% and 4% alternatives remain unqualified assumptions.

Response domain	Current monitor	Proposed future dose intervals
Four moving single-route cases	Reject	Finite response-domain pass
Two isolated paired-route cases	Reject	Finite response-domain pass
Repeated and changed-context/fault cases	Reject	Reject

These admissions are response-component checks. Baseline error, reference availability, execution timing and online freshness still have to fit the complete correction gate. A continuous sparse policy cannot discard old pulse effects after 600 ms: no release rule beyond the measured response window has been qualified.

The biological transduction is explicitly reduced: charge-normalized drive is held for its 25 ms frame after a chosen 35 ms onset. The experiment identifies that surrogate; it does not prove how real spinal tissue responds to a 500 microsecond biphasic waveform.

The connected reset-trial pilot

The new protocol allows at most one whole intervention per reset three-second trial. A declared washout premise separates trials while learned coefficients carry. Sixteen independent captures test four plasticity families at four starting coefficient scales. The common forecast context fails: its 150 ms allocation reaches 24.964 mm against the unchanged 15 mm limit. All 128 practice trials withhold stimulation; all 64 retention/transfer assessments stay OFF. No arm advantage or learning multiplier is shown.

Expert reference uncertainty comes from the maximum independent complete-trajectory 2*RMSE, about 3.12 mm, rather than a training fit. Runtime qualification also binds the tested physical support and exact observed past pulse pattern. Passing a context name alone grants no authority beyond that evidence.

The delayed-residual guard checks targets only after their measurements arrive. Timed checks also expire missing observations when no later packet arrives. Cancellation stops continuation and cannot undo a delivered pulse.

06 / Turn model limits into requirements

Four fresh observed steady-response calibrations recover the current maps before varying prospective monitor uncertainty. The measured response floor remains fixed. No hidden recruitment matrix or threshold enters the requirement calculation.

Comparison allocation	Full single-route precision requirement	Meaning
8 mm forecast input	5.06–6.64% of nominal phase charge	A proposed 4% interval leaves scalar room in these calibrations.
15 mm forecast input	0.354–0.471% of nominal phase charge	Spending the forecast budget sharply tightens delivery precision.
Two full simultaneous routes	Fails even at zero charge uncertainty	The 19 mm response floor plus forecast error already exceeds 25 mm.

These are steady effective-map requirements, not permission for fractional physical pulses or evidence that a new circuit has achieved the stated precision. The 8 mm forecast input is a comparison target; it is not a newly demonstrated assisted forecast. The dynamic single-frame study supplies a separate profile and response domain.

Detailed electronics must earn the allocation

Investigate independently evidenced charge integration or edge capture. Bound both phases, gain and offset, bandwidth and intersample waveform error, integration-window edges, reset/hold behavior, correlated errors and simultaneous sources. A current threshold or increased nominal ADC rate is not a complete-waveform bound.

Stage 4 then implements native multichannel KiCad circuits, programmable routing, monitoring, power/telemetry and embedded timing. Stage 5 populates parametric CAD from actual boards, components, package and lead interfaces. Stage 2 allocation solids are preliminary resource models, not that completed assembly.

The pressure branch remains a second target-specific mechanism: external pressure feedback with identified positive pressor authority, disturbances and finite stop/hold/ramp fallbacks. Dense-stimulation neural recording remains unavailable in that model. It does not supply arbitrary pressure lowering or universal body automation.

07 / Inspect, reproduce and extend

The repository separates assumptions, observed synthetic records, qualification gates, proposals and biological unknowns. Historical results remain unchanged. Saved-evidence audits bind model sources and artifacts; separate runners recompute the experiments. The Git manifest binds committed blobs independently of checkout line endings.

Entry point	Purpose
CURRENT_STATUS.md	Current stage, implemented scope and remaining integration.
docs/stage3/README.md	Technical dossier and experiment-specific reproduction commands.
scripts/verify_stage3_revision.py	Saved revision evidence, source identities and rejection-boundary audit.
docs/stage3/isolated-pilot.md	Connected protocol, conservative drift rejection and pulse-free OFF results.
docs/stage3/mechanism-to-electronics-handoff.md	Concrete detailed-design requirements and unresolved gates.

Selected primary references

[1] El Hadwe et al. (2026). Circumferential spinal recording and movement decoding.

[2] McPherson et al. (2015). Activity-dependent spinal stimulation during retraining.

[3] Nishimura et al. (2013). Corticospinal plasticity with targeted stimulation.

[4] Phillips et al. (2025). Spinal stimulation for blood-pressure regulation.

[5] Formento et al. (2018). Proprioceptive consequences of spinal stimulation.

AstroSam / Mind-Link: design-format inspiration.

Reference descriptions are shortened topic labels. Cohorts, applicability and full source records are preserved in the Stage 1 audit and the evidence-to-design revision. The original 28-page hypothesis paper remains read-only and is not an original experimental result.

Contribution record. Pratham defines the project ambition and scope. Codex/ChatGPT assists literature checking, modelling, implementation, review, figures and writing. Detailed decision and assistance records are in provenance/CONTRIBUTIONS.md. This report is an inspectable digital revision, not a claim of independent human efficacy.